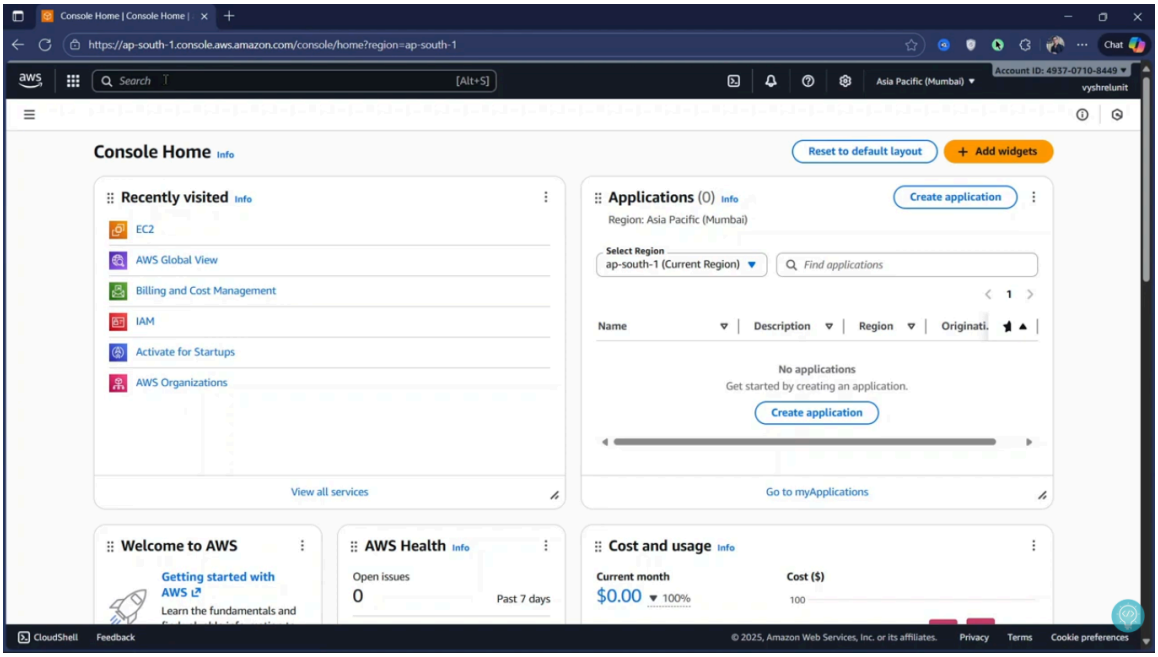
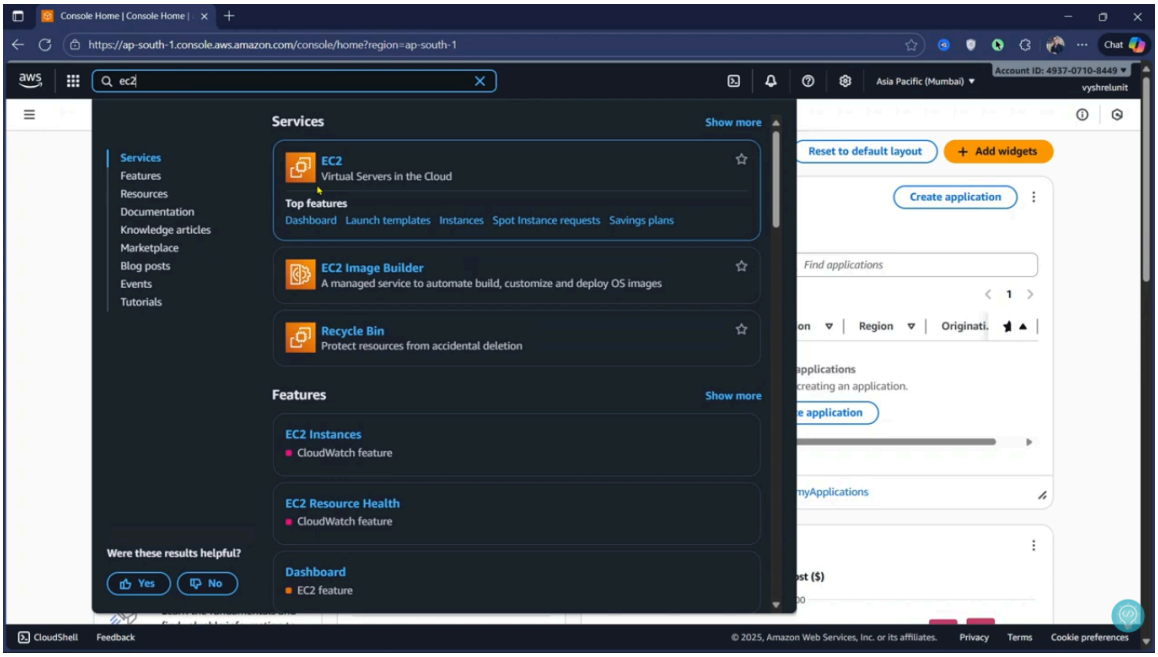
Car Price Prediction Project Deployment on AWS EC2 (Step-by-Step Guide)

1. Create AWS Account





1. Go to https://aws.amazon.com

2. Click "Create AWS Account"

3. Enter Email, Password, and AWS account name

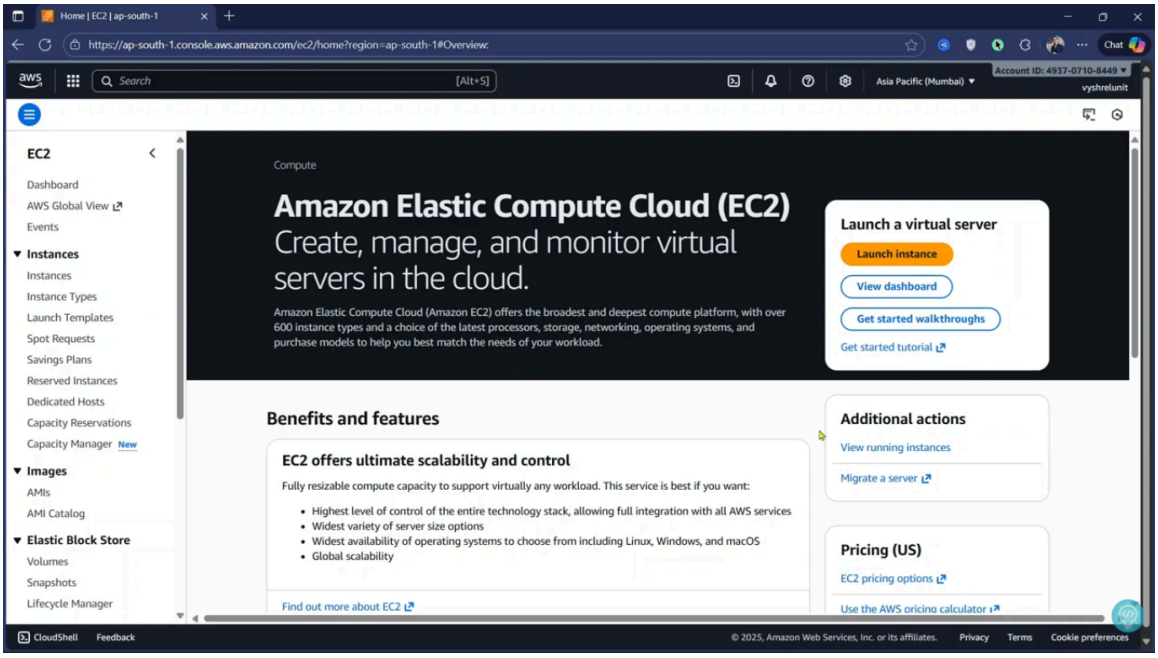
4. Add payment details (required even for free tier)

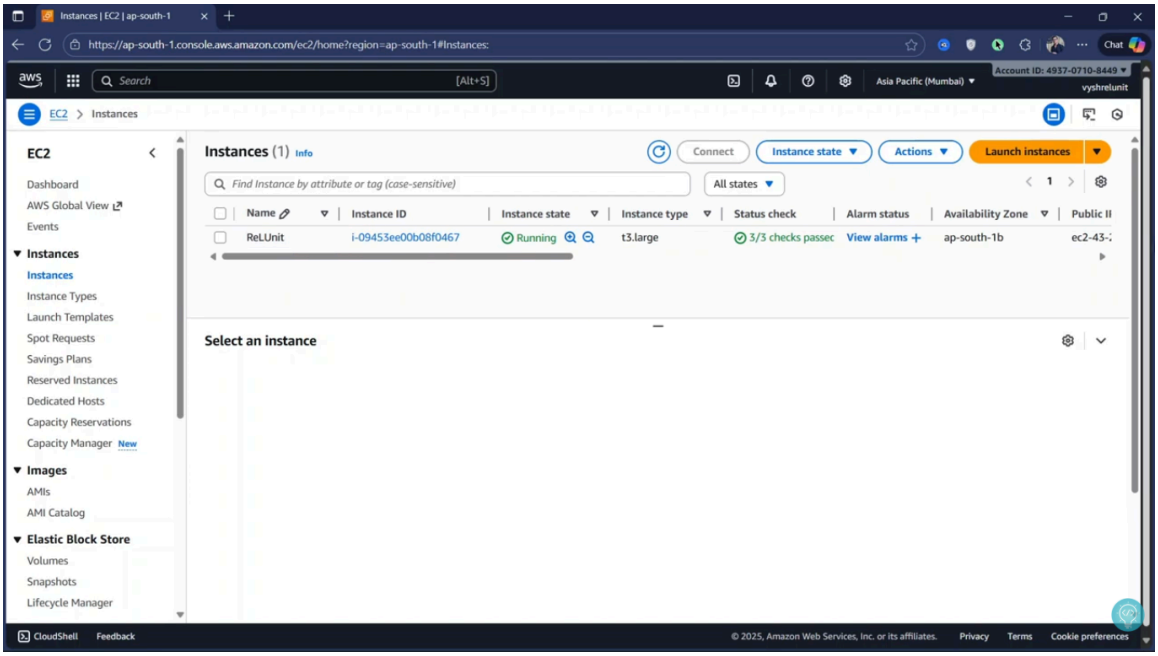
5. Verify phone number

6. Choose Basic Support Plan (Free)

7. Login to AWS Management Console

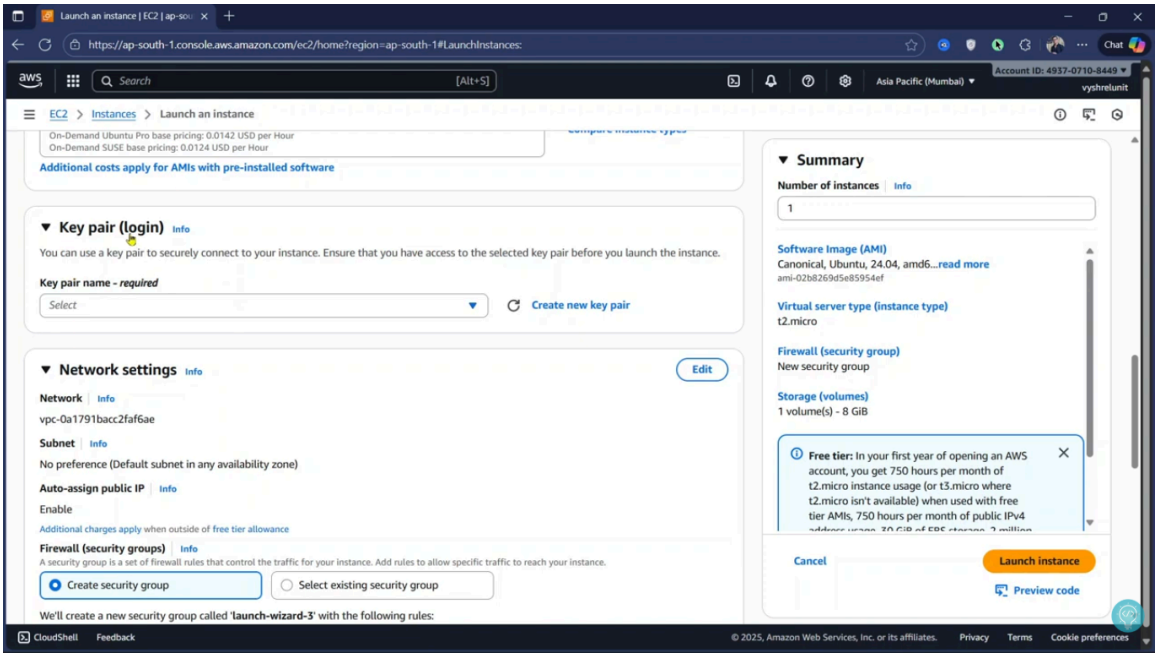
2. Launch an EC2 Instance





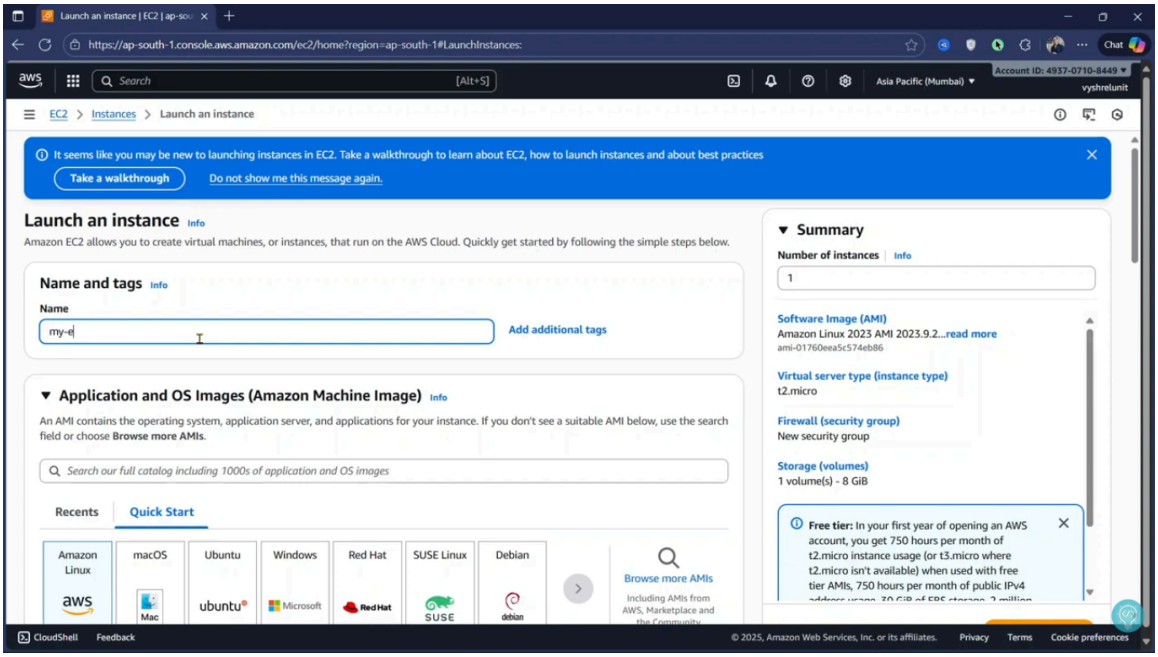
1. In AWS Console search for EC2
2. Click "Launch Instance"
3. Name: Car-Price-Prediction-Server
4. Choose AMI: Ubuntu Server 22.04 LTS
5. Instance type: t2.micro (Free tier eligible)

3. Create Key Pair



1. Click "Create Key Pair"
2. Key pair name: carprice-key
3. Type: RSA
4. Format: .pem
5. Download and save the file safely (used to connect to server)

4. Configure Network Settings



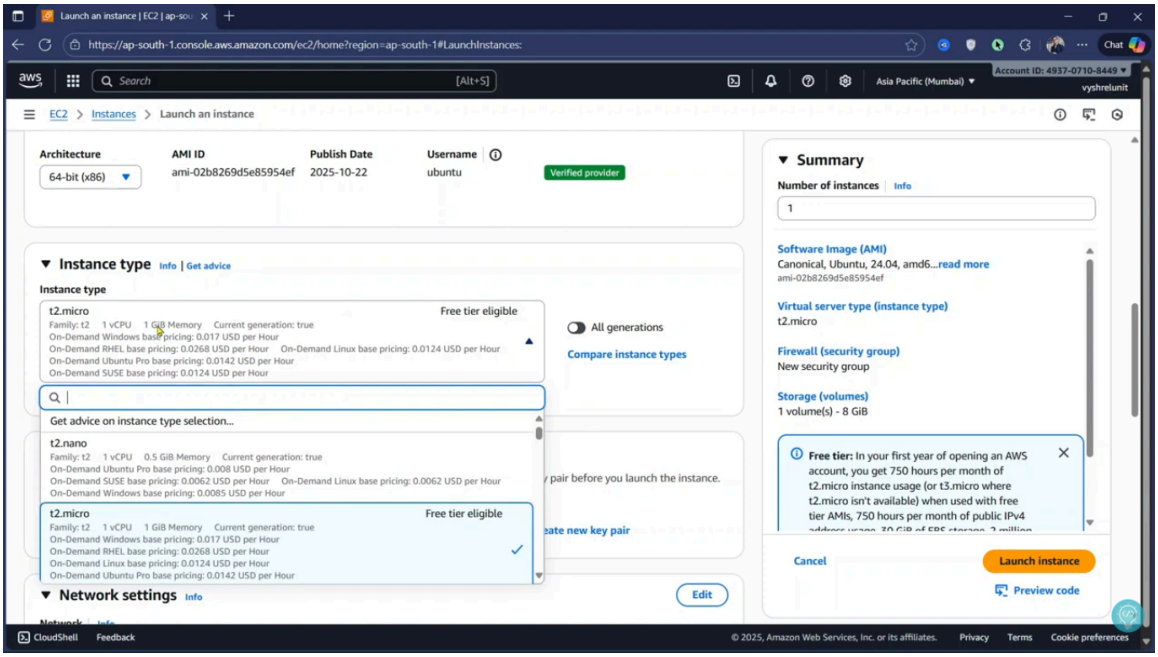
Allow the following ports:

• SSH – Port 22

• HTTP – Port 80

• Custom TCP – Port 5000 (for Flask application)

5. Storage Configuration



Leave the default storage setting:

• 8 GB General Purpose SSD

6. Launch Instance

Click "Launch Instance"

Your EC2 server will start running.

7. Connect to EC2 Server

1. Go to Instances in EC2 Dashboard

2. Copy the Public IPv4 Address

3. Open Git Bash or Terminal

4. Navigate to the folder containing the .pem file

5. Run:

ssh -i carprice-key.pem ubuntu@YOUR_PUBLIC_IP

Example:

ssh -i carprice-key.pem ubuntu@3.110.125.20

8. Update the Server

Run the following command:

sudo apt update

9. Install Required Software

Install Python and pip:

sudo apt install python3-pip -y

10. Upload Project Files Using WinSCP

1. Open WinSCP

2. Host name: Public IP of EC2

3. Username: ubuntu

4. Private key: carprice-key.pem

Upload the following files:

• main.py

• index.html

• random_forest_regression_model.pkl

11. Install Required Python Libraries

Run:

pip3 install flask

pip3 install numpy

pip3 install pandas

pip3 install scikit-learn

12. Modify Flask App for External Access

Ensure the following line exists in main.py:

app.run(host='0.0.0.0', port=5000)

13. Run the Application

Execute:

python3 main.py

14. Access the Web Application

Open a browser and go to:

http://YOUR_PUBLIC_IP:5000

Example:

http://3.110.125.20:5000

Your Car Price Prediction web application will be visible.

15. Architecture Overview

User Browser

↓

Public IP

↓

AWS EC2 Server

↓

Flask Application

↓

Random Forest Model (.pkl)

↓

Predicted Car Price